Meet the microcontroller

IoT · Level 2 · Lesson 4



Today you will

- Say what a microcontroller is and where you find one.
- Explain a program that reads a sensor, decides and acts, again and again.
- Write a night-light program as blocks or a flowchart.

Read, decide, act

- A computer on one chip
- Inputs in, outputs out
- Set up once, then loop forever
- You choose the threshold

Night light, three ways

1. Draw the flowchart
2. Act it out with a torch
3. Build it in MakeCode
4. Test it with the light slider
5. Choose your threshold

Power the micro:bit from USB or its own battery pack only, never from mains. Ask before you use a family phone or computer.

Let's discuss

1. Which devices around you run a loop like this?
2. Why is it useful that the threshold is a number you can change?
3. What could go wrong if the light sensor sat right next to the LED?

Check yourself

- What happens to a night-light program with no loop?